

Explicit formulas for the Segal–Shale–Weil representation: Schwartz models, Bruhat 2-cocycles, multipliers

Compiled from Rao (1993), Kashiwara–Vergne (1978),
Kudla (1996), Lion–Vergne (1980), Blüher (1996), and successors

1 Setting and notation

Throughout, F is a local field of characteristic $\neq 2$ (so $F = \mathbb{R}, \mathbb{C}$, or a non-archimedean local field of residue characteristic $\neq 2$); the formulas below are stated in the form that uniformly covers all three cases except where flagged. Fix a nontrivial additive character $\psi : F \rightarrow \mathbb{C}^\times$.

Let $(W, \langle \cdot, \cdot \rangle)$ be a $2n$ -dimensional symplectic space over F , with a complete polarisation $W = X \oplus Y$ into transversal Lagrangians. With respect to this decomposition write elements of $\mathrm{Sp}(W)$ as $g = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$, where $a : X \rightarrow X$, $b : X \rightarrow Y$, $c : Y \rightarrow X$, $d : Y \rightarrow Y$ satisfy the symplectic relations.

The Siegel parabolic $P = P_Y$ is the stabiliser of Y :

$$P = \left\{ \begin{pmatrix} a & b \\ 0 & (a^\vee)^{-1} \end{pmatrix} : a \in \mathrm{GL}(X), \ b a^{-1} = (b a^{-1})^t \right\}.$$

Its Levi is $M = \{m(a) := \mathrm{diag}(a, (a^\vee)^{-1}) : a \in \mathrm{GL}(X)\}$ and its unipotent radical is $N = \{n(b) := \begin{pmatrix} 1 & b \\ 0 & 1 \end{pmatrix} : b \in \mathrm{Hom}(X, Y), \ b = b^t\}$. Bruhat decomposition: $\mathrm{Sp}(W) = \bigsqcup_{0 \leq j \leq n} P w_j P$, where $w_j = \mathrm{diag}(w'_j, I_{2(n-j)})$ and w_n is the long Weyl element exchanging the two halves.

By $\mathcal{S}(X)$ we denote the Bruhat–Schwartz space (Schwartz functions in the archimedean case; locally constant compactly supported functions in the non-archimedean case).

Set $\mathrm{Mp}(W)$ for the metaplectic two-fold cover of $\mathrm{Sp}(W)$ when $F = \mathbb{R}$; in the non-archimedean case the same notation refers to the unique nontrivial topological double cover constructed via the cocycle below [Weil1964].

2 Schrödinger model: explicit operators on $\mathcal{S}(X)$

The Stone–von Neumann theorem produces, up to a scalar, a unique unitary action of $\mathrm{Sp}(W)$ on $\mathcal{S}(X)$ implementing the symplectic action on the Heisenberg group. The resulting projective representation is the *Segal–Shale–Weil* (or oscillator, or Weil) representation. We denote any choice of lift by $r = r_\psi$ and the resulting cocycle by c , so that

$$r(g_1) r(g_2) = c(g_1, g_2) r(g_1 g_2), \quad |c| = 1. \tag{1}$$

The standard Schrödinger formulas for elements of P and for the long Weyl element w_n are (Kudla [?, Ch. II, Prop. 2.3]; Rao [?, §3]; Lion–Vergne [LV1980]):

$$r(m(a))\phi(x) = |\det a|^{1/2} \phi(xa), \quad (2)$$

$$r(n(b))\phi(x) = \psi\left(\frac{1}{2}\langle x, xb \rangle\right) \phi(x), \quad (3)$$

$$r(w_n)\phi(x) = \gamma \widehat{\phi}(x) = \gamma \int_Y \psi(-\langle x, y \rangle) \phi(y) dy, \quad (4)$$

where dy on Y is the ψ -self-dual Haar measure and $\gamma = \gamma(\psi \circ q_X)^{-1}$ is the inverse Weil index attached to the standard quadratic form on X (Section 3). For a partial Weyl element w_j (j rank), Eq. (4) becomes a partial Fourier transform on the corresponding j coordinates; see [?, §3] and [?, Ch. I].

Closed form for a general element

For $g = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in \mathrm{Sp}(W)$, with $\mathrm{rk}(c) = j$, Rao [?, Lemma 3.2] and Kudla [?, Prop. 2.3] give

$$r(g)\phi(x) = \int_{\ker(c) \setminus Y} \psi\left(\frac{1}{2}\langle xa, xb \rangle - \langle xb, yc \rangle + \frac{1}{2}\langle yc, yd \rangle\right) \phi(xa + yc) d\mu_g(y), \quad (5)$$

where $d\mu_g$ is the unique Haar measure on $\ker(c) \setminus Y$ making $r(g)$ an isometry of $L^2(X)$. The integrand is a quadratic phase; on P (where $c = 0$) it reduces to Eqs. (2) to (3), and on w_n (where c is invertible) it reduces to Eq. (4).

Equivalently (Rao’s ”intertwining” form): set $f_g(x, y^*) = \psi(\text{quadratic form attached to } g)$; then $r(g)$ is an integral operator with kernel built from f_g and a Haar measure μ_g on the quotient $Y^*/\ker \gamma_g$, normalised so that r is unitary [?, Thm. 3.5].

3 The Weil index of a quadratic character

Definition 3.1 (Weil index, [Weil1964].) Let (V, B) be a non-degenerate quadratic space over F with associated form $q(v) = \frac{1}{2}B(v, v)$, and let $\phi_q(v) = \psi(q(v))$. Choose Haar measures on V and its ψ -dual V^* that are mutually ψ -dual. Then there exists a constant $\gamma(\psi \circ q) = \gamma(\phi_q) \in \mathbb{C}$ with $|\gamma| = 1$ such that, as tempered distributions,

$$\widehat{\phi_q dv} = \gamma(\phi_q) |q|^{-1/2} \phi_q^{-1} dv^*.$$

The Weil index has the following key properties:

1. $\gamma(\psi \circ q)$ depends only on the isometry class of q [?, §14].
2. It is multiplicative on orthogonal direct sums: $\gamma(\psi \circ (q_1 \perp q_2)) = \gamma(\psi \circ q_1) \gamma(\psi \circ q_2)$.
3. It is an 8-th root of unity when $\dim V$ is rational, and explicitly, for the diagonal form $q = \sum a_i x_i^2$ in m variables,

$$\gamma(\psi \circ q) = \gamma(\det q, \psi) \gamma(\psi)^m \varepsilon(q), \quad \varepsilon(q) = \prod_{i < j} (a_i, a_j)_F, \quad (6)$$

where $(\cdot, \cdot)_F$ is the Hilbert symbol and $\gamma(a, \psi)$ is the relative Weil index $\gamma(\psi \circ ax^2)/\gamma(\psi \circ x^2)$, valued in $\{\pm 1, \pm i\}$.

Lemma 3.2 (Modified Weil-index identity; [Kudla1996]. , Ch. I] For $b \in \text{Hom}(X, Y)$ symmetric and invertible, and $f \in \mathcal{S}(X)$,

$$\int_Y \int_X \psi\left(\frac{1}{2}\langle x, xb \rangle + \langle x, y \rangle - \frac{1}{2}\langle y, y b^{-1} \rangle\right) f(xb) dx dy = \gamma(\phi_b) |b|^{-1/2} \int_Y f(y) dy.$$

This identity is the engine that converts the formal manipulation Eq. (5) into a unitary operator and that produces the cocycle of Section 4 as a product of Weil indices.

4 The Rao cocycle relative to Bruhat decomposition

Fix the Lagrangian Y . The non-trivial cocycle attached to the section r above admits a clean closed form in terms of the Weil index of the *Leray invariant* of a triple of Lagrangians.

Definition 4.1 (Leray invariant). Given an ordered triple (Y_0, Y_1, Y_2) of Lagrangians in W , in suitably generic position there exists a symmetric $b \in \text{Hom}(Y_1, Y_0)$ with $Y_2 = Y_1 n(b)$. The isometry class of the quadratic form $q_b(y) = \frac{1}{2}\langle y, yb \rangle$ on Y_1 is independent of all auxiliary choices, and is called the *Leray invariant* $L(Y_0, Y_1, Y_2) \in W(F)$ (Witt group).

Theorem 4.2 (Rao, [Rao1993]. Thm. 4.1] With the section r defined by Eqs. (2) to (4), Eq. (5) and the standard normalisation [?, Thm. 3.5],

$$c^Y(g_1, g_2) = \gamma(\psi \circ L(Y, Y g_1^{-1}, Y g_2 g_1^{-1}))^{-1}. \quad (7)$$

Equivalently, $c^Y(g_1, g_2) = \gamma(\psi \circ L(Y, Y g_1^{-1}, Y(g_1 g_2)^{-1} g_1^{-1}))^{-1}$ in the conventions of Kudla [?, Thm. 3.1].

The cocycle satisfies (Rao [?, §4]):

1. *P-bi-invariance.* $c(p_1 g_1 p'_1, p_2^{-1} g_2 p_2) = c(g_1, g_2)$ for $p_1, p'_1, p_2 \in P$.
2. *Vanishing on Weyl pairs.* $c(g_1, g_2) = 1$ whenever $g_1 g_2 \in W^P$ (the Weyl group representatives).
3. *Inversion.* $c(g, g^{-1}) = 1$ [?, Cor. 4.2] since the Leray invariant is then trivial.
4. *Tensor compatibility.* For block-diagonal pairs $g = \text{diag}(g_S)$ and g' ,

$$c(g, g') = \prod_S c_S(g_S, g'_S).$$

Big-cell special case (recovers Weil's original formula)

If $g_1 = w_n u_1 p_1$ and $g_2 = w_n u_2 p_2$ both lie in the open Bruhat cell $P w_n P$, formula Eq. (7) reduces to

$$c(g_1, g_2) = \gamma\left(\psi \circ q_{u_1^{-1} + u_2}\right),$$

which is exactly the cocycle written by Weil [?, Thm. 3, p. 164] on the big cell.

Two-dimensional ($\text{Sp}_2 = \text{SL}_2$) case

For $g_j = \begin{pmatrix} a_j & b_j \\ c_j & d_j \end{pmatrix} \in \text{SL}_2(F)$, Rao [?, Cor. 4.3] gives

$$c(g_1, g_2) = \begin{cases} 1, & \text{if } b_1 = 0, b_2 = 0, \text{ or } c_1 c_2 \in (F^\times)^2, \\ \gamma_F(c_1 c_2 (c_1 a_2 + d_1 c_2) b_1 b_2), & \text{otherwise.} \end{cases} \quad (8)$$

For $F = \mathbb{R}$ this agrees with the Kubota cocycle $c^K(g_1, g_2) = (x(g_1), x(g_1 g_2)/x(g_2))_{\mathbb{R}}$ where $x(g) = c$ if $c \neq 0$ and $x(g) = d$ otherwise; cf. Blüher [?, Prop. 10(c)].

5 Normalised metaplectic cocycle and multiplier

To pass from the $\mathbb{C}^1$ -valued cocycle c to a $\{\pm 1\}$ -valued metaplectic 2-cocycle one introduces a coboundary correction.

Definition 5.1 (Rao multiplier, [Rao1993]. Def. 5.2] For $g \in \mathrm{Sp}(W)$ in the Bruhat cell $\Omega_j = Pw_jP$, define

$$x(g) \in F^\times / (F^\times)^2, \quad x(p_1 w_j p_2) := \det(p_1 p_2|_Y) \pmod{(F^\times)^2}.$$

This is well defined: $x(w_j) = 1$, $x(m(a)) = \det(a)$. The *Rao multiplier* is

$$m(g) = \frac{\gamma_F(x(g), \eta)}{\gamma_F(\eta)^{j(g)}}, \quad \eta := \psi^{1/2} \text{ (formal)}, \quad (9)$$

i.e., $\beta_{e,\psi}(g) = \gamma(x(g), \eta)^{-1} \gamma(\eta)^{-j(g)}$ in Kudla's notation.

Theorem 5.2 (Normalised cocycle, [Rao1993]. Thm. 5.3] Setting $\tilde{r}(g) := m(g) r(g)$, the resulting cocycle $\tilde{c}$ takes values in $\{\pm 1\}$. Explicitly, with $g_1 \in \Omega_{j_1}$, $g_2 \in \Omega_{j_2}$, $g_1 g_2 \in \Omega_j$, and Leray invariant q on a space of dimension ℓ , set $t = j_1 + j_2 - j - \ell$. Then

$$\tilde{c}(g_1, g_2) = (x(g_1), x(g_2))_F (-x(g_1)x(g_2), x(g_1 g_2))_F ((-1)^t, \det q)_F (-1, -1)_F^{t(t-1)/2} h_F(q), \quad (10)$$

where $h_F(q)$ is the Hasse–Witt invariant of q and $(\cdot, \cdot)_F$ is the Hilbert symbol.

Special evaluations [?, Cor. 5.4–5.5]:

$$\tilde{c}(g, g^{-1}) = (x(g), (-1)^j x(g))_F (-1, -1)_F^{j/2}, \quad (11)$$

$$\tilde{c}(w_{S_1}, w_{S_2}) = (-1, -1)_F^{|S_1 \cap S_2|(|S_1 \cap S_2|+1)/2}. \quad (12)$$

Over a finite field or over $\mathbb{C}$ all these symbols vanish: $\tilde{c} \equiv 1$, recovering the linearity of the Weil representation in those cases.

6 Lion–Vergne form: cocycle as a Maslov index

A choice-free reformulation, due to Lion and Vergne, expresses c on $\mathbb{R}$ as an exponentiated triple Maslov / Kashiwara index of Lagrangians.

Definition 6.1 (Kashiwara triple index). For an ordered triple (ℓ_1, ℓ_2, ℓ_3) of Lagrangians in $(W, \langle \cdot, \cdot \rangle)$, the symmetric form on $\ell_1 \oplus \ell_2 \oplus \ell_3$ defined by

$$q_B(x_1 + x_2 + x_3) = \langle x_1, x_2 \rangle + \langle x_2, x_3 \rangle + \langle x_3, x_1 \rangle$$

has, over $\mathbb{R}$, a well-defined signature $\tau(\ell_1, \ell_2, \ell_3) := \mathrm{sgn}(q_B) \in \mathbb{Z}$. Over a p -adic field, τ takes values in the Witt group $W(F_p)$.

Theorem 6.2 (Lion–Vergne, [LV1980]. / Over $F = \mathbb{R}$, fix a Lagrangian ℓ_0 . Then

$$c(g_1, g_2) = \exp\left(-\frac{i\pi}{4} \tau(\ell_0, \ell_0 g_1, \ell_0 g_2 g_1)\right). \quad (13)$$

Over a p -adic field of residue characteristic $\neq 2$,

$$c_p(g_1, g_2) = \gamma_p(\tau_p(\ell_0, \ell_0 g_1, \ell_0 g_2 g_1))^{-1}, \quad (14)$$

where $\gamma_p : W(F_p) \rightarrow \mathbb{C}^\times$ is a fixed character of the Witt group; in either case the resulting class lies in $H^2(\mathrm{Sp}(W); \mathbb{Z}/2\mathbb{Z})$.

The relation between the Lion–Vergne cocycle and Rao’s expression is precisely that the Leray invariant $L(\ell_0, \ell_0 g_1^{-1}, \ell_0 g_2 g_1^{-1})$ is the class of the form q_B in $W(F)$; the two formulations are isomorphic encodings of the same $\mathrm{Sp}(W)$ -equivariant function on triples of Lagrangians.

7 Maximal-compact splitting and the Bluher cocycle

Bluher [Bluher1996] constructs an explicit section $r^+ : \mathrm{Sp}(2m, \mathbb{R}) \rightarrow \widetilde{\mathrm{Sp}}(2m, \mathbb{R})$ whose cocycle c^+ admits a closed form on the maximal compact subgroup. Write $K = U(m) \hookrightarrow \mathrm{Sp}(2m, \mathbb{R})$ via $u(A + iB) = \begin{pmatrix} A & B \\ -B & A \end{pmatrix}$.

Theorem 7.1 ([Bluher1996]., Prop. 10] *Let $k \in K$ have eigenvalues $e^{i\theta_1}, \dots, e^{i\theta_m}$ with $-\pi < \theta_j \leq \pi$, and set $\theta = \sum \theta_j$, $\delta = \sum \delta_j$ where*

$$\delta_j = \begin{cases} -1, & \theta_j \in (-\pi, 0), \\ 0, & \theta_j = 0, \\ 1, & \theta_j \in (0, \pi), \\ 2, & \theta_j = \pi. \end{cases}$$

Then $r^+(k) = e^{-i\theta m/4} e^{i\pi\delta/2} k(k)$, where $k : K \rightarrow \widetilde{\mathrm{Sp}}$ is the canonical compact splitting characterised by $k(k)\Phi_0 = \Phi_0$ on the Gaussian $\Phi_0(x) = e^{-\pi\|x\|^2}$. The associated cocycle satisfies

$$c^+(pg_1, g_2) = c^+(g_1, g_2) = c^+(g_1, g_2 p) \quad \text{for all } p \in P^+ = \{p \in P : \det(a) > 0\},$$

and for $m = 1$ coincides with the Kubota cocycle on $\mathrm{SL}_2(\mathbb{R})$.

Proposition 7.2 ([Bluher1996]., Prop. 2] *Explicit lifts to $\widetilde{\mathrm{Sp}}(2m, F)$:*

$$(\det A, -1)_F^{1/2} a(A) \in \widetilde{\mathrm{Sp}}, \quad n(B) \in \widetilde{\mathrm{Sp}}, \quad \gamma(1)T_j \in \widetilde{\mathrm{Sp}}.$$

8 Comparative summary

Variant	Cocycle / multiplier	Values	Locus where simplest
Weil (1964) original	$c(g_1, g_2) = \gamma(\psi \circ q_{u_1^{-1} + u_2})$ on $Pw_n P$ only	$\mathbb{C}^1$ (8th roots of phase)	Big Bruhat cell
Rao (1993)	$c^Y(g_1, g_2) = \gamma(\psi \circ L(Y, Y g_1^{-1}, Y g_2 g_1^{-1}))^{-1}$	$\mathbb{C}^1$	All cells; P -biinvariant
Rao normalised	$\tilde{c}$ via multiplier $m(g) = \gamma(x(g), \eta) \gamma(\eta)^{-j(g)}$, Eq. (10)	$\{\pm 1\}$ formula	All Bruhat cells
Kashiwara index / Lion–Vergne (1980)	$c(g_1, g_2) = \exp(-\frac{i\pi}{4} \tau(\ell_0, \ell_0 g_1, \ell_0 g_2 g_1)) \gamma_p(\tau_p)^{-1}$ (p -adic)	$= \mathbb{Z}/8\mathbb{Z} \hookrightarrow \mathbb{C}^1$, in fact $\mathbb{Z}/2\mathbb{Z}$ ($\mathbb{R}$);	Choice-free in Lagrangians

Variant	Cocycle / multiplier	Values	Locus where simplest
Kubota / SL_2 form	$c^K(g_1, g_2)$ $(x(g_1), x(g_1 g_2)/x(g_2))_F$	$= \{\pm 1\}$	$n = 1$ only
Bluher (1996) r^+	$r^+(k) = e^{-i\theta m/4} e^{i\pi\delta/2} k(k)$ on K ; $r^+(p_1 w p_2) = r_P(p_1) \varepsilon(w) r_P(p_2)$	$\{\pm 1\}$	Maximal compact K
Kashiwara–Vergne (1978)	Action by $\mathfrak{sp}(W)$ via degree- ≤ 2 differential operators (oscillator); decomposition under reductive dual pairs (G, G')	Lie-algebra	Fock model; harmonic decomposition
Thomas (2008)	Character $\Theta(g)$ as a Weil index of a Maslov form, choice-free; pulled back from $\mathrm{Mp}(W \oplus W)$	$\mathbb{C}$ on conjugacy classes	Character computations

9 The Kashiwara–Vergne harmonic-polynomial framework

A complementary, infinitesimal viewpoint is provided by Kashiwara and Vergne [KV1978]. In the Fock (Bargmann) model of the Schrödinger representation, the symplectic Lie algebra $\mathfrak{sp}(W)$ acts by polynomials of degree ≤ 2 in creation and annihilation operators $a_i^\dagger, a_i$ satisfying $[a_i, a_j^\dagger] = \delta_{ij}$. Quadratic monomials in $a^\dagger a^\dagger, aa, a^\dagger a$ generate the (complexified) symplectic algebra and integrate, via the metaplectic cover, to the Segal–Shale–Weil representation.

The principal results of [KV1978] are:

1. For a reductive dual pair (G, G') in $\mathrm{Sp}(W)$ with G compact, the restriction of the oscillator representation to $\widetilde{G} \times \widetilde{G'}$ decomposes multiplicity-freely:

$$\omega \cong \bigoplus_{\sigma \in \widehat{G}} \sigma \otimes \theta(\sigma),$$

with each $\theta(\sigma)$ an irreducible unitarisable highest-weight module of $\widetilde{G'}$.

2. For the dual pair $(O(p, q), \mathrm{Sp}(2n, \mathbb{R}))$ the irreducible constituents of the oscillator representation, restricted to the compact factor, are realised in spaces of *harmonic polynomials* on the polarised vector space — polynomials annihilated by all G -invariant constant-coefficient differential operators on the relevant Lagrangian — and these spaces classify the unitary highest-weight representations of the non-compact partner.
3. The list of unitarisable highest-weight modules of $\mathfrak{su}(n, n)$ obtained this way is exactly the *Kashiwara–Vergne list* [KV1978, Jakobsen2021], parametrised by Young diagrams of width $\leq n - 1$.

The Kashiwara–Vergne realisation is dual to the Schrödinger picture of Section 2: it works on the Lie-algebra level inside the Fock model, while the Rao / Lion–Vergne formulas work at the group level. The two are reconciled by the Cayley transform mapping the Schrödinger and Fock models, whose explicit kernel is itself a Weil-index Gaussian.

10 Recent activity (post-1990 highlights)

- **Rao 1993** [Rao1993]: the canonical reference for explicit Schwartz-model formulas and the closed-form $\mathbb{C}^1$ and $\{\pm 1\}$ cocycles.
- **Bluher 1996** [Bluher1996]: explicit section r^+ on $\mathrm{Sp}(2m, F)$ and its compatibility with the Kubota cocycle.
- **Kudla 1994 / 1996** [Kudla1994, Kudla1996]: splittings of metaplectic covers of dual reductive pairs, and a streamlined exposition of the Schwartz-model action.
- **Thomas 2008** [Thomas2008]: choice-free character formula for the Weil representation, expressed via Weil index of a Maslov-type form on $W \oplus W$.
- **Pan 2001** [Pan2001]: explicit splitting formulas for metaplectic covers restricted to several reductive dual pairs.
- **Zemel 2012** [Zemel2015]: p -adic computation of the Weil representation of discriminant forms, free of theta-function inputs.
- **Prasad 2009** [Prasad2009]: simple algebraic re-derivation of character/decomposition formulas for finite-abelian-group Weil representations.
- **Jakobsen 2021** [Jakobsen2021]: reconstruction of the Kashiwara–Vergne list of unitary highest-weight modules of $\mathfrak{su}(n, n)$ via Weyl-algebra homomorphisms; quantum analogue.
- **Trias 2026** [Trias2026]: ℓ -modular Stone–von Neumann theorem, modular Weil representation and an analogue of Rao’s metaplectic cocycle in characteristic $\ell \neq p$, including the surprising splitting at $\ell = 2$.
- **Wang 2025–2026** [Wang2025, Wang2026]: Siegel modular forms from the Weil representation, with explicit 8-th roots of unity arising from comparing Rao, Kudla, Perrin, Lion–Vergne and Satake–Takase 2-cocycles.
- **Wei Li 2024** [LiWenWei2024]: behaviour of the metaplectic cocycle and endoscopic transfer factors under variation of the additive character ψ .

11 Variants of the cocycle and how they are related

The various 2-cocycles in the literature differ by coboundaries that fall into two classes:

1. *Choice of section*, i.e. choice of distinguished representatives in each Bruhat cell. Different sections produce cocycles differing by a coboundary $\partial f(g_1, g_2) = f(g_1)f(g_2)/f(g_1g_2)$ where f is the relative phase between the two sections. This is how Rao’s c and Bluher’s c^+ are related.
2. *Choice of polarisation/Lagrangian*. Different Lagrangians Y, Y' produce cocycles $c^Y, c^{Y'}$ that differ by an explicit coboundary built from a partial Fourier transform; Kudla [?, Ch. II] writes this coboundary in terms of $\beta_{e, \psi}$ and μ_V .

The cohomology class is a single element of $H^2(\mathrm{Sp}(W); \mathbb{C}^1)$ (or, after normalisation, of $H^2(\mathrm{Sp}(W); \mathbb{Z}/2\mathbb{Z})$ in the local non-archimedean and real cases) — this class is intrinsic; the formulas above are explicit cocycle representatives differing by coboundaries.

12 References

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